SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	LAU YEE WEN
IC No. / Matric No.	A23CS0099
Year / Program	2/SECPH
Section	01
Lecturer Name	Ms. Rozilawati
Case Study	WBL Project
System Name	Koperasi KADA Online System
Group Name	Technicrab

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

The Employee Module of the Koperasi KADA Online System is intended to address a number of issues inherent in the current manual registration and application processes. The module has two main features: applying for membership and viewing application status. Prospective members can provide their personal information via an easy-to-use online interface that includes built-in validation checks to ensure data accuracy. The system automates application processing, reducing delays and providing real-time status updates. Members can also track the status of their applications via the online portal, which provides clear updates like "Pending," "Approved," or "Rejected." This transition from a manual to a digital process eliminates inefficiencies like time-consuming paperwork, the storage of sensitive data in physical formats, and delays caused by manual verification.

A number of issues with the current manual process have prompted the transition to an online system. These include an inefficient and labour-intensive registration process that requires manual data entry and poses security risks due to the storage of personal information in physical formats. Members also have difficulty determining the status of their applications, relying on phone calls or emails for updates, which is both inefficient and prone to delays. The current system's heavy reliance on human resources for data entry, verification, and communication increases the risk of errors and puts a strain on staff. Furthermore, the manual system results in high error rates, processing delays, and difficulties in managing physical documents, all of which disrupt service delivery.

Finally, there are security concerns about storing sensitive customer information in physical formats, which exposes it to unauthorised access, loss, or damage. This combination of inefficiencies and security risks highlights the importance of digitising and streamlining these processes using the Koperasi KADA Online System.

Group Member's Name	Matric No	Module
Lau Yee Wen	A23CS0099	Employee Module (Apply Membership & View Membership Application Status)
Cheryl Cheong Kah Voon	A23CS0060	Member module (View financial status & Manage profile)

Chau Ying Jia	A23CS0213	Member module (Apply for loan & View loan application status)
Neo Li Xin	A23CS0253	Admin module (Generate reports)
Woo Cheng Shuan	A23CS0283	Admin module (Approve membership registration & loan application)

1.2 Project Weaknesses or Improvement

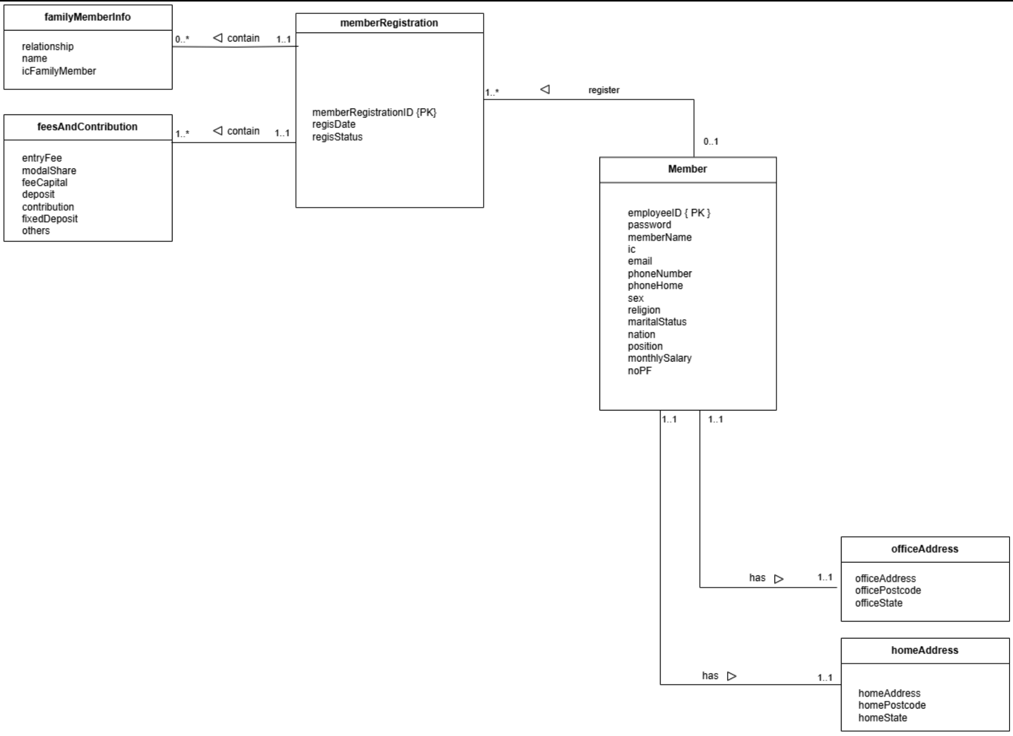
The original system includes modules like "Apply Membership" and "View Membership Application Status," which enable members to submit applications and track their progress. However, a significant flaw in the proposed system is the inability for members to update or correct their application forms after submission. This limitation may result in inaccurate or outdated information being stored in the system, especially if members make mistakes when filling out forms or their personal information changes. As a result, members must rely on administrators to manually handle corrections, increasing administrative workload and causing application processing delays.

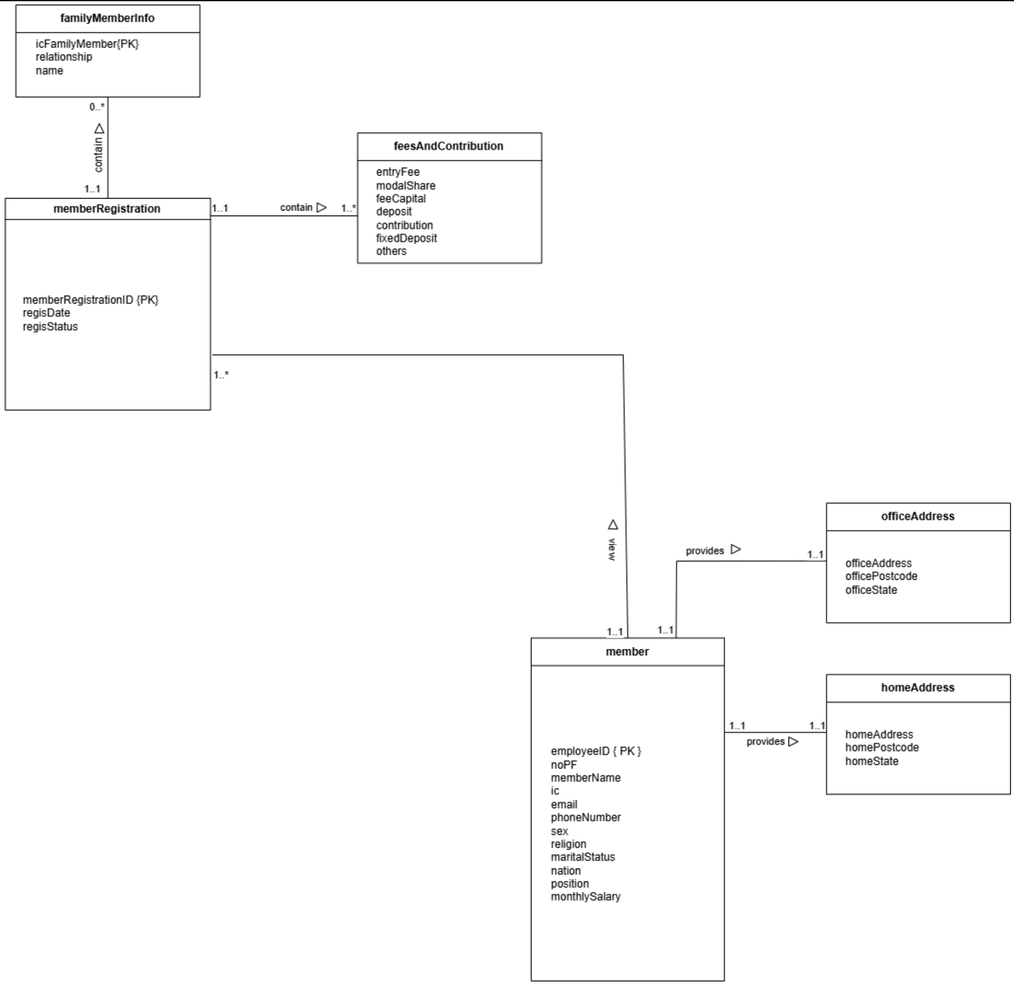
1.3 Proposed Module

To address this issue, a new module, "Edit Member Application Form," has been proposed. This module allows members to make changes to their submitted application forms, but only if their application status is still "Pending." The system will automatically detect the application status, and if it is "Approved" or "Rejected," editing will be disabled. This ensures data integrity and prevents tampering with finalised applications.

The "Edit Member Application Form" module also includes notification features that let members know the status of their applications and whether they are eligible to make changes. Administrators will receive alerts when changes are made to submitted forms, allowing them to quickly review and verify the updated information.

By incorporating this module, the system becomes more user-friendly, allowing members to correct errors or update their information without requiring administrative assistance. It also reduces administrators' workload, ensures accurate data collection, and improves the overall efficiency of the membership application process.

SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	**Conceptual ERD**	
		
2.2	**Logical ERD**	



2.3 Data Dictionary for Logical ERD

Entity	Attribute	Data Type	Constraints	Description
Member	-employeeID	INT(4)	Primary Key, NOT NULL	Unique identifier for each member
	-memberName	VARCHAR(255)	NOT NULL	Name of the member
	-email	VARCHAR(255)	Unique, NOT NULL	Member’s email address
	-phoneNumber	VARCHAR (20)	NOT NULL	Primary contact number

			-ic	VARCHAR(20)	NOT NULL	Identity card number.
			-maritalStatus	VARCHAR(20)	NOT NULL	Marital status (e.g., Single, Married).
			-sex	VARCHAR(10)	NOT NULL	Gender of the member.
			-religion	VARCHAR(20)	NOT NULL	Religion of the member.
			-nation	VARCHAR(50)	NOT NULL	Nationality of the member.
			-no_pf	VARCHAR(50)	NOT NULL	Job-related number (e.g., EPF No.).
			-position	VARCHAR(100)	NOT NULL	Job position of the member.
			-phoneHome	VARCHAR(20)	NOT NULL	Home contact number.
			-monthlySalary	DECIMAL(10,2)	NOT NULL	Monthly salary amount.
		memberRegistration	-memberRegistrationID	INT(11)	Primary Key, NOT NULL	Primary key for the table.
			-regisDate	DATE	NOT NULL	Date when the registration was made.
			regisStatus	VARCHAR(20)	NOT NULL	Registration status (e.g.,

						Pending, Approved).
		feesAnd Contribut ion	-entryFee	INT(11)	NOT NULL	Initial registration fee.
			-modalShare	INT(11)	NOT NULL	Member's share of the capital.
			feeCapital	INT(11)	NOT NULL	Contribution to the capital fund.
			-deposit	INT(11)	NOT NULL	Deposit amount for the membership.
			-contribution	INT(11)	NOT NULL	Contribution s made by the member.
			-fixedDeposit	INT(11)	NOT NULL	Amount kept in fixed deposit.
		familyM emberInf o	-icFamilyMe mber	VARCHAR(20)	Primary Key, NOT NULL	
			-relationship	VARCHAR(50)	NOT NULL	Relationship to the primary member
			-name	VARCHAR(100)	NOT NULL	Full name of the family member
		officeAd dress	-officeAddre ss	TEXT	NOT NULL	Full office address.

		-officePostcode	VARCHAR(10)	NOT NULL	Postcode for the office address.
		-officeState	VARCHAR(20)	NOT NULL	State where the office is located.
	homeAddress	-homeAddress	TEXT	NOT NULL	Full home address.
		-homePostcode	VARCHAR(10)	NOT NULL	Postcode for the home address.
		-homeState	VARCHAR(20)	NOT NULL	State where the home is located.

2.4	**Relational Database Schema**

One to One

Member (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary)
PK: employeeID

feesAndContribution (employeeID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit)
PK: employeeID
FK: employeeID references Member(employeeID)

memberRegistration (memberRegistrationID, employeeID, regisDate, regisStatus)
PK: memberRegistrationID
FK: employeeID references Member(employeeID)

familyMemberInfo (icFamilyMember, employeeID, relationship, name)
PK: icFamilyMember, employeeID
FK: employeeID references Member (employeeID)

homeAddress(employeeID, homeAddress, homePostcode, homeState)
PK: employeeID
FK: employeeID references Member (employeeID)

officeAddress(employeeID, officeAddress, officePostcode, officeState)

		PK: employeeID FK: employeeID references Member (employeeID)
2.5	**Normalization**	
		1. memberRegistration- familyMemberInfo 1NF: memberRegistration_familyMemberInfo (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary, icFamilyMember, relationship, name) PK: employeeID, icFamilyMember 2NF: Member (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) PK: employeeID familyMemberInfo (icFamilyMember, relationship, name, employeeID) PK: icFamilyMember FK: employeeID references member (employeeID) 3NF: Member (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) PK: employeeID familyMemberInfo (icFamilyMember, relationship, name, employeeID) PK: icFamilyMember FK: employeeID references member (employeeID) 2. memberRegistration-feesAndContribution 1NF: memberRegistration(memberRegistrationID, regDate, regStatus, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit) PK: memberRegistrationID 2NF: memberRegistration(memberRegistrationID, regDate, regStatus) PK: memberRegistrationID feeAndContribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, memberRegistrationID) PK: memberRegistrationID

FK: memberRegistrationID references memberRegistration (memberRegistrationID)

3NF:
memberRegistration(memberRegistrationID, regDate, regStatus)
PK: memberRegistrationID
feeAndContribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, memberRegistrationID)
PK: memberRegistrationID
FK: memberRegistrationID references memberRegistration (memberRegistrationID)

3. memberRegistration-feesAndContribution

1NF:
memberRegistration(memberRegistrationID, regDate, regStatus, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit)
PK: memberRegistrationID

2NF:
memberRegistration(memberRegistrationID, regDate, regStatus)
PK: memberRegistrationID

feeAndContribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, memberRegistrationID)
PK: memberRegistrationID
FK: memberRegistrationID references memberRegistration (memberRegistrationID)

3NF:
memberRegistration(memberRegistrationID, regDate, regStatus)
PK: memberRegistrationID

feeAndContribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, memberRegistrationID)
PK: memberRegistrationID
FK: memberRegistrationID references memberRegistration (memberRegistrationID)

4. member-memberRegistration

1NF:
Member_memberRegistration(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary,memberRegistrationID, regDate, regStatus)
PK: employeeID

		2NF: Member (<u>employeeID</u>, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) PK: employeeID memberRegistration(memberRegistrationID, regDate, regStatus, employeeID) PK: memberRegistrationID FK: employeeID references memberRegistration (memberRegistrationID) 3NF: Member (<u>employeeID</u>, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) PK: employeeID memberRegistration(<u>memberRegistrationID</u>, regDate, regStatus, employeeID) PK: memberRegistrationID FK: employeeID references memberRegistration (memberRegistrationID) 5. Member - homeAddress 1NF: Member_homeAddress(<u>employeeID</u>, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary, homeAddress, homeState, homePostcode) PK: employeeID 2NF: Member (<u>employeeID</u>, no_pf, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) PK: employeeID homeAddress (<u>employeeID</u>, homeAddress, homePostcode, homeState) PK: employeeID FK: employeeID references Member(employeeID)

3NF:
Member (employeeID, no_pf, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary)
PK: employeeID

homeAddress (employeeID, homeAddress, homePostcode, homeState)
PK: employeeID
FK: employeeID references Member(employeeID)

6. Member - officeAddress

1NF:
Member_homeAddress(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary, homeAddress, homeState, homePostcode)
PK: employeeID

2NF:

Member (employeeID, no_pf, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary)
PK: employeeID

officeAddress (employeeID, officeAddress, officePostcode, officeState)
PK: employeeID
FK: employeeID references Member(employeeID

3NF:
Member (employeeID, no_pf, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary)
PK: employeeID

officeAddress (employeeID, officeAddress, officePostcode, officeState)
PK: employeeID
FK: employeeID references Member(employeeID)

Normalized Relations:

		1. Member (employeeID, no_pf, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary) 2. familyMemberInfo (icFamilyMember, relationship, name, employeeID) 3.memberRegistration(memberRegistrationID, regDate, regStatus, employeeID) 4. feeAndContribution(memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit) 5. homeAddress (employeeID, homeAddress, homePostcode, homeState) 6. officeAddress (employeeID, officeAddress, officePostcode, officeState)